

Exercise 11

Q1. Develop a movie theatre booking simulator that manages a 3×3 seating grid layout. The program should accept user row and column selections and change that seat's status keyword from "O" (Open) to "X" (Reserved).

```
print("Movie Theatre Booking Simulator")
```

```
seats = [  
    ["O", "O", "O"],  
    ["O", "O", "O"],  
    ["O", "O", "O"]  
]
```

```
print("Initial Seating Layout:")
```

```
for row in seats:
```

```
    print(row)
```

```
row = int(input("Enter row number (1-3): "))
```

```
col = int(input("Enter column number (1-3): "))
```

```
if 1 <= row <= 3 and 1 <= col <= 3:
```

```
    if seats[row - 1][col - 1] == "O":
```

```
        seats[row - 1][col - 1] = "X"
```

```
        print("Seat reserved successfully!")
```

```
    else:
```

```
        print("Seat is already reserved.")
```

else:

```
print("Invalid row or column.")
```

```
print("\nUpdated Seating Layout:")
```

for row in seats:

```
print(row)
```

Output:

```
PS C:\Users\DELL\Documents\Shubhada Shinde> c:: cd 'c:\Users\DELL\Documents\Shubhada Shinde'; & 'C:\Users\DELL\AppData\Local\Microsoft\WindowsApps\python\pythoncore-3.14-64\python.exe' 'c:\Users\DELL\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\lib\python\pythoncore-3.14-64\python.exe' 'c:\Users\DELL\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\lib\python\pythoncore-3.14-64\python.exe' 'c:\Users\DELL\Documents\Shubhada Shinde\exercise 11.1.py'
Movie Theatre Booking Simulator
Initial Seating Layout:
['O', 'O', 'O']
['O', 'O', 'O']
['O', 'O', 'O']
Enter row number (1-3): 2
Enter column number (1-3): 3
Seat reserved successfully!

Updated Seating Layout:
['O', 'O', 'O']
['O', 'O', 'X']
['O', 'O', 'O']
PS C:\Users\DELL\Documents\Shubhada Shinde> 
```

Q2. Create a daily class schedule grid where each row represents an hour slot and columns correspond to days of the week, allowing users to view or overwrite subject topics dynamically.

```
print("Daily Class Schedule Grid")
```

```
days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]
```

```
schedule = [
```

```
    ["Maths", "Python", "OS", "DBMS", "AI"],
```

```
    ["Python", "Maths", "DBMS", "AI", "OS"],
```

```
["OS", "AI", "Maths", "Python", "DBMS"],  
["AI", "DBMS", "Python", "OS", "Maths"]  
]
```

```
print("Class Schedule:")
```

```
for i in range(4):
```

```
    print("Hour", i + 1, ":", schedule[i])
```

```
# Overwrite a subject
```

```
hour = int(input("\nEnter hour slot (1-4): "))
```

```
day = int(input("Enter day (1-5): "))
```

```
subject = input("Enter new subject: ")
```

```
if 1 <= hour <= 4 and 1 <= day <= 5:
```

```
    schedule[hour - 1][day - 1] = subject
```

```
    print("\nSchedule updated successfully!")
```

```
else:
```

```
    print("Invalid hour or day.")
```

```
# Display updated schedule
```

```
print("\nUpdated Class Schedule:")
```

```
for i in range(4):
```

```
    print("Hour", i + 1, ":", schedule[i])
```

Output:

```
PS C:\Users\DELL\Documents\Shubhada Shinde> Open folder in new window (ctrl + click) ents\Shubhada Shinde'; & 'C:\Users\DELL\AppData\Local\Microsoft\Windows\apps\python\pythoncore-3.14-64\python.exe' 'c:\Users\DELL\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy-launcher' '56658' '--' 'C:\Users\DELL\Documents\Shubhada Shinde\exercise 11.2.py'
Daily Class Schedule Grid
Class Schedule:
Hour 1 : ['Maths', 'Python', 'OS', 'DBMS', 'AI']
Hour 2 : ['Python', 'Maths', 'DBMS', 'AI', 'OS']
Hour 3 : ['OS', 'AI', 'Maths', 'Python', 'DBMS']
Hour 4 : ['AI', 'DBMS', 'Python', 'OS', 'Maths']

Enter hour slot (1-4): 2
Enter day (1-5): 3
Enter new subject: C++

Schedule updated successfully!

Updated Class Schedule:
Hour 1 : ['Maths', 'Python', 'OS', 'DBMS', 'AI']
Hour 2 : ['Python', 'Maths', 'C++', 'AI', 'OS']
Hour 3 : ['OS', 'AI', 'Maths', 'Python', 'DBMS']
Hour 4 : ['AI', 'DBMS', 'Python', 'OS', 'Maths']
PS C:\Users\DELL\Documents\Shubhada Shinde> 
```